

• Nothing is uploaded

UniLab

Select a file → transform it on this device → download the result.

A student's file jobs — compress, convert, scan, trim — done entirely inside the browser, so the file never leaves the device and no deletion promise has to be trusted.

15

students interviewed, 22 Aug – 8 Sep, fifteen programmes

12

pains recorded — six we predicted, six we did not

all

requirements trace to a stated pain; two we built are parked, in the open

Myo Zin Thant
Product Owner

Swan Htut Oakkar Aung
Designer

Thiha Lin
Tech Lead

Wanna San
AI Lead

Zaw Win Htut
QA / Test

One workflow, fifty-eight settings.

1 · SELECT

Pick a file on this device

Phone camera roll, Files, a drag onto the page. Type and size screened on every path.



2 · TRANSFORM ON THIS DEVICE

The whole job runs in the tab

pdf.js · WebCodecs · Tesseract · RNNoise, all in the browser. **No upload endpoint exists in the codebase.**



3 · DOWNLOAD

Take the result — and watch it expire

Held in memory only, 30:00 countdown, *Delete now*, purge on tab close. Proven by test.

58 tools are 58 *settings* of that one workflow — not 58 workflows. Split PDF and Remove Noise differ in what they compute, not in what the student does.

PDF 20

Image 13

Audio 11

Video 7

Everyday 3

Text 2

Study 2

— eighteen video and audio settings: the part no free PDF site has at all.

Fifteen people, fifteen programmes.

Zaw Win Htut

S1

Y2 Nursing Science
EN · chat · phone → laptop

S6

Y3 Cosmetic Science
TH · chat · phone → laptop

S11

Y1 English
TH · chat · phone → library
PC

Myo Zin Thant

S2

Y3 Software Engineering
EN · in person · laptop only

S7

ID DOC

Y1 Traditional Chinese Medicine
EN · chat · phone only

S12

ID DOC

Y4 Digital Tech. for Business Innovation
EN · chat · copy shop → phone

Wanna San

S3

Y1 Business Administration
TH · chat · phone → laptop

S8

Y4 Accounting
EN · in person · laptop only

S13

Y2 Tourism Business & Events
EN · call · phone → laptop

Swan Htut Oakkar Aung

S4

ID DOC

Y4 Public Health
MY · chat · phone only

S9

Y2 Aviation Business Mgmt
EN · chat · phone only

S14

ID DOC

Y3 Environmental Health
EN · chat · phone → laptop

Thiha Lin

S5

Y2 Laws
EN · call · iPad → laptop

S10

Y3 Multimedia Tech. & Animation
MY · chat · phone only

S15

Y1 Applied Chemistry
TH · chat · phone only

WHEN

22 Aug – 8 Sep 2026 · 10–18 min each

WHO

MFU undergraduates · Y1 4 · Y2 4 · Y3 4 · Y4 3 · **nobody from this course**

CHANNEL

chat 10 · in person 2 · voice call 3

LANGUAGE

EN 9 · TH 4 · MY 2 — quotes kept in the original

DEVICE

phone only 5 · phone → laptop 7 · laptop only 2 · shop scanner → phone 1

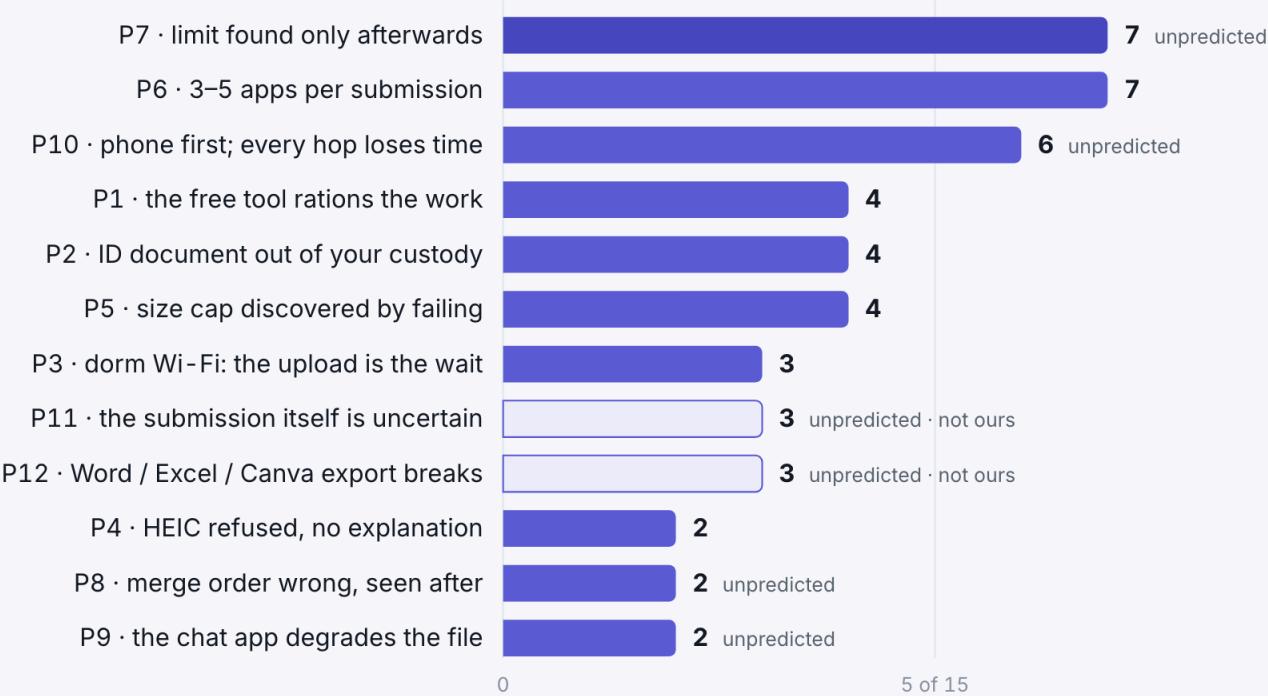
IDENTITY DOC

4 of 15 — passport, medical certificate, Thai ID, signed permission sheet

RECORDED

role · pain · quote. **No names, no student IDs** — anywhere (LR7)

What they said.



7 of 15

found out about the limit only after the work was done. "Upload failed" — no reason. "File type not supported" — no fix. The pain no hypothesis predicted, and the one our estimate-before-the-run already answers.

6

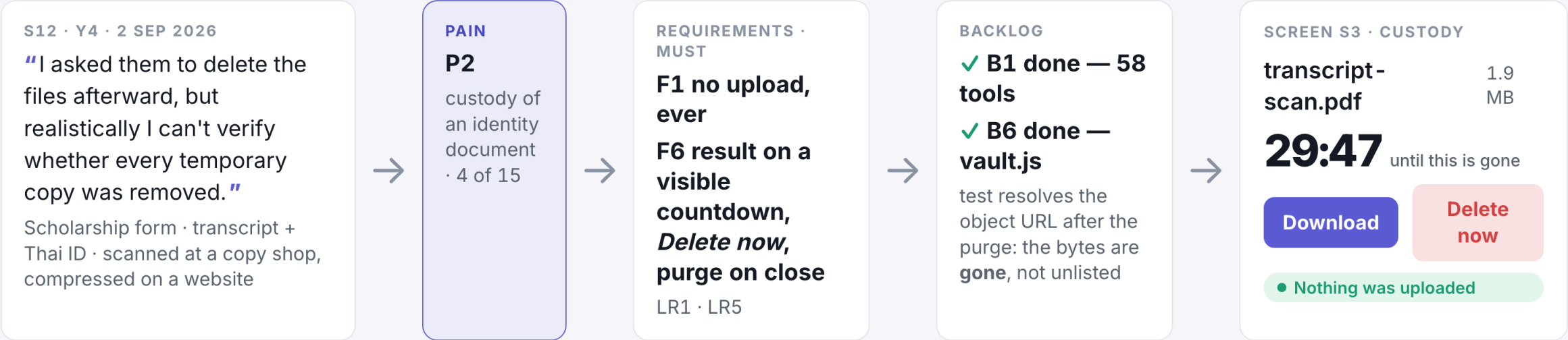
pains nobody predicted. Six hypotheses became pains; six more pains arrived on their own, including the largest.

2

recorded and not ours. The LMS itself, and Word's export — written down as boundaries, not hidden.

The pain is rarely "I can't make a PDF." It's the workflow around it — and the limit you meet afterwards.

One thread, six documents.



Said in one breath: **a student cannot verify a deletion promise** (W1) → so no file may leave the device (W2, LR1) → so the result lives in memory with a countdown she can watch (W3, F6) → so screen S3 shows **29:47** and **Delete now** (W4). Every screen in the prototype traces backwards like this.

What we got wrong.

- H1 Free tools ration the work; the cap lands at the deadline → P1 · 4 of 15
- H2 An ID document uploaded is in someone else's custody → P2 · 4 of 15
- H3 On campus Wi-Fi the upload is part of the wait → P3 · 3 — still untimed
- H4 iPhone HEIC photos get rejected → P4 · 2 of 15
- H5 Size caps force compression against a specific number → P5 · 4 — number usually unknown
- H6 Thai / Burmese scans are unsearchable, so quotes get retyped 0 of 15 → F8 parked
- H7 No one free site covers documents, video and audio → P6 · 7 of 15
- H8 The same multi-step chore repeats every week 0 of 15, contradicted → F11 parked
- H9 "Free for students" is not a differentiator confirmed · nobody switches for free

2

features we had already built. Nobody asked. OCR in nineteen languages, saved Workflows. Not re-justified — **parked:** out of the requirement table, code untouched, decision written down for week six.

0 of 15

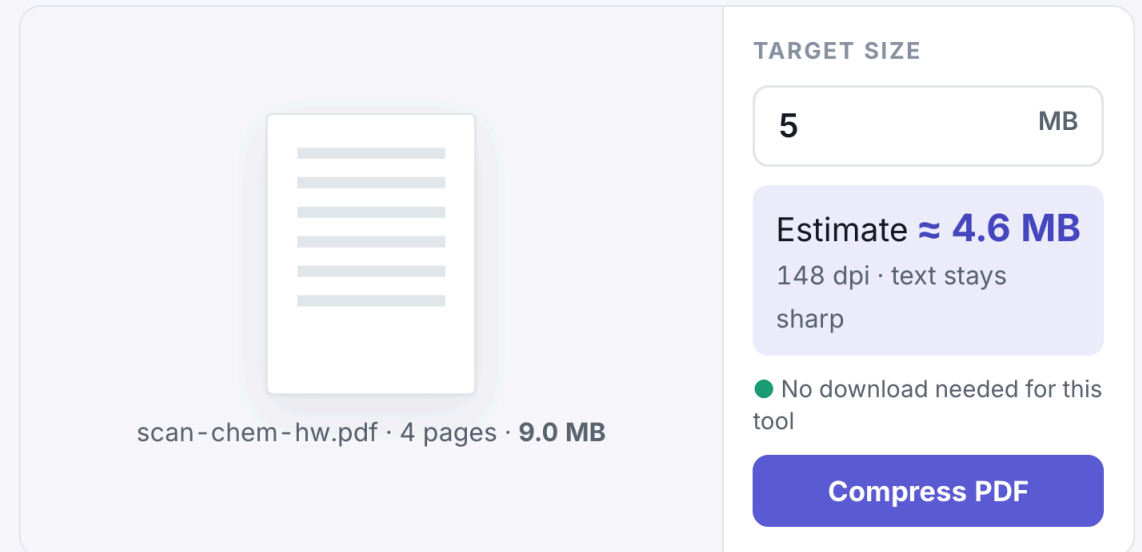
named "free" as a reason to switch. What they named: fewer steps in one place (S1, S13), no ads (S3), and local processing for identity documents (S4, S14).

Every requirement still in the spec traces to a stated pain — because we took two out.

It runs with the Wi-Fi off.

1. **Show the airplane-mode icon.** "This laptop has no connection."
2. **Compress PDF** is already open with a scanned 9 MB PDF — "the size S15 got refused with last week."
3. Type the cap: **5 MB**. Read the estimate: **≈ 4.6 MB, before anything runs.** "That's F5 — and that's P7 answered."
4. **Run. Download.** Then press **Delete now** — "same purge the countdown does."
5. Close: **"Nothing left this machine, because nothing could. There is no upload endpoint in the codebase."**

Before the room, while online: open Compress PDF and drop the file on it, *then* switch to airplane mode. The tool module is lazy-loaded; loaded once, it runs offline. Offline first = a spinner, not a demo. If anything fails on stage, show the mock-up on the right and say the live product is at thiha-lynn.github.io/unilab. Do not debug in front of the room.



Prototype screen S2 (mock, from design-system.md tokens only) — the estimate sits above the button, before the run. R1: one primary action. R2: 16 px input on phones.

Compliance is architecture.

- LR1** No user file content leaves the device; no upload endpoint exists — PDPA minimisation
- LR2** No account, email, name or phone for any tool
- LR3** Network fetches carry program code or model data only, disclosed before they start — 2 of 2 tools verified
- LR4** No "files never touch a server" claim unless the architecture makes it true — it does, so the privacy page reads its numbers from the code
- LR5** Results in memory only, expire unaided, erasable on demand, unreachable once dropped — **proven by test**
- LR6** Redaction, blurring and EXIF stripping remove content, never cover it — EXIF proven; redaction test still to write (B15)
- LR7** No interview PII into an AI tool or the public repo — role, pain, quote only, kept for all 15
- LR8** No filename or file content reaches the console in production

A duty you have discharged can fail. A duty you never incurred cannot.

CCA §26 access log, 90 days, a launch blocker	We store nothing for anyone. No server, no session, no IP — there is no event to log.
Consent banner, ad-script gating, DPA	No ads, no analytics, no third party collects anything through the page.
ETA §9/26 consent records	Nothing is transmitted, so nothing creates the obligation.

Written conditionally, not deleted: *if the system ever stores a file for a user, this duty returns in full.* The class draft that assumed a server is preserved verbatim beside the rewrite.

What is still open.

B17 · TOPIC

UniLab is not on the 19 Aug assigned list

The course overview says pick your own; the slide lists five. The confirmation request is written and goes to you today — with a mapped fallback if the answer is "take one of the five".

P3 · A NUMBER

Campus Wi-Fi round trip: perception, not yet a measurement

Three participants said the upload is the wait. UniLab's half is measured — zero network attempts across five exit APIs. The competitor's half needs campus Wi-Fi, a 10 MB fixture and three runs.

B11 · THE CENTRAL CLAIM

"No file bytes leave the device" — tested on one chain, not all 58

Statically clean across the codebase; dynamically proven on compress → images-to-PDF. The per-tool sweep is the first BUILD-phase test.

Q5 · Q6 · PARKED

OCR and saved Workflows — re-admit or cut, in week six

Zero of fifteen raised either. Three options each are written in the spec; the OCR branch gets asked directly in the next five interviews.

15 interviewed · **12** pains · every requirement traced · **2** parked, in the open.

Questions.

github.com/Thiha-Lynn/unilab · start at .docs/01–
requirements/TRACEABILITY.md · every participant at 02–
interviews/results.md

"Tell me about one of them." — **whoever ran it**: role, date, task, the pain in their words, the quote. Zaw S1 S6 S11 · Myo S2 S7 S12 · Wanna S3 S8 S13 · Swan S4 S9 S14 · Thiha S5 S10 S15

"Where are the names?" — **Zaw**: nowhere, by rule LR7. Public repo. The code-to-person map was never written down

"Where's the OCR?" — **Zaw**: shipped, parked. Zero of fifteen asked. Spec §2.1, Q5

"Biggest surprise?" — **Wanna**: P7. Nobody was stopped by a PDF; seven were stopped by finding out too late

"Why isn't UniLab on the topic list?" — **Myo**: overview says pick your own; confirming today; fallback mapped to Internship Management System, rule.md and the tests transfer

"Fifty-eight tools in a month?" — **Myo**: fifty-eight settings of one workflow; the core feature is what month two must ship end to end

"Every requirement traces to a pain?" — **Thiha**: every F, NFR and LR — the spec has the column. Two didn't, so they're out

"Is free your pitch?" — **Myo**: no. One place, fewer steps; custody for identity documents